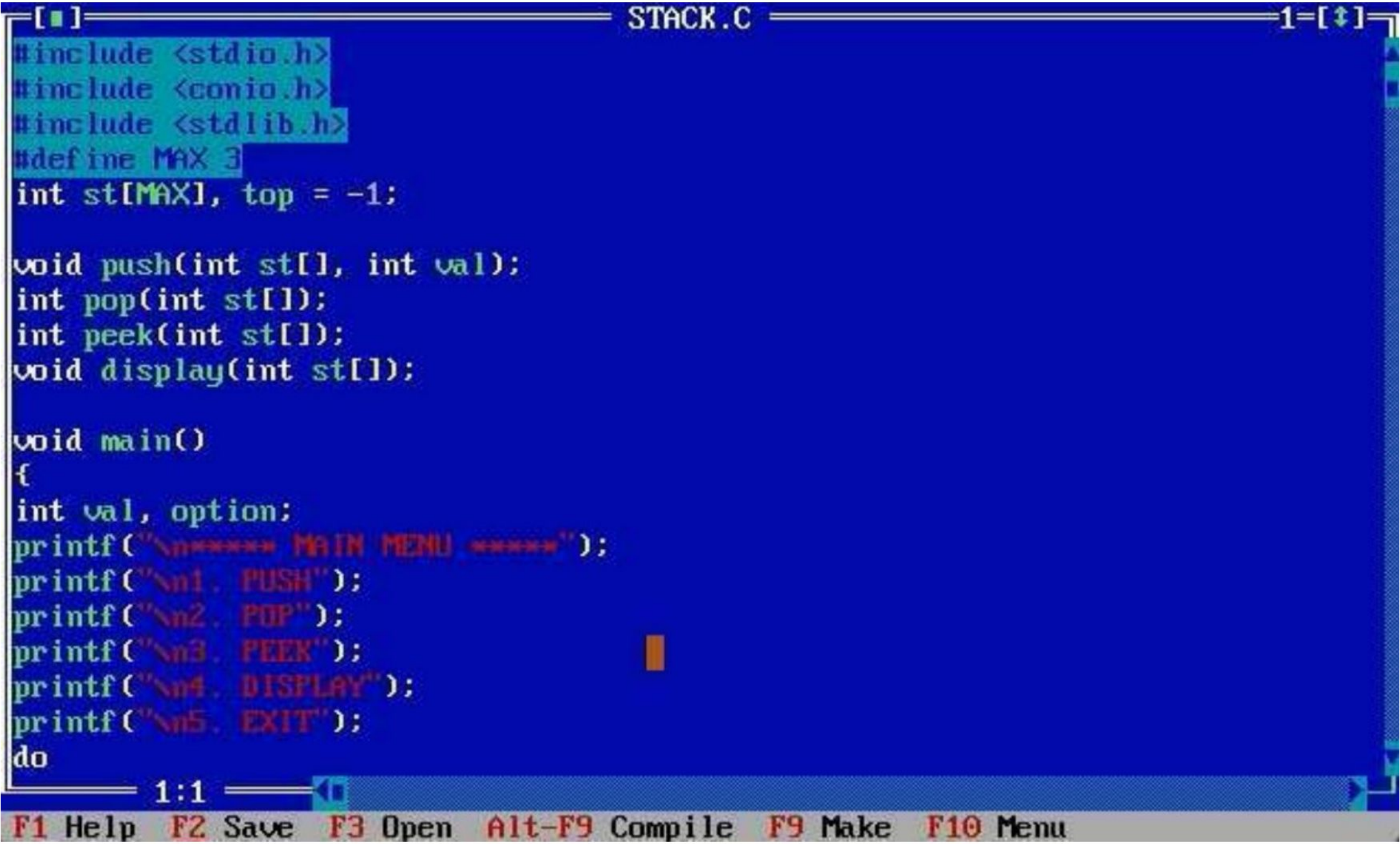
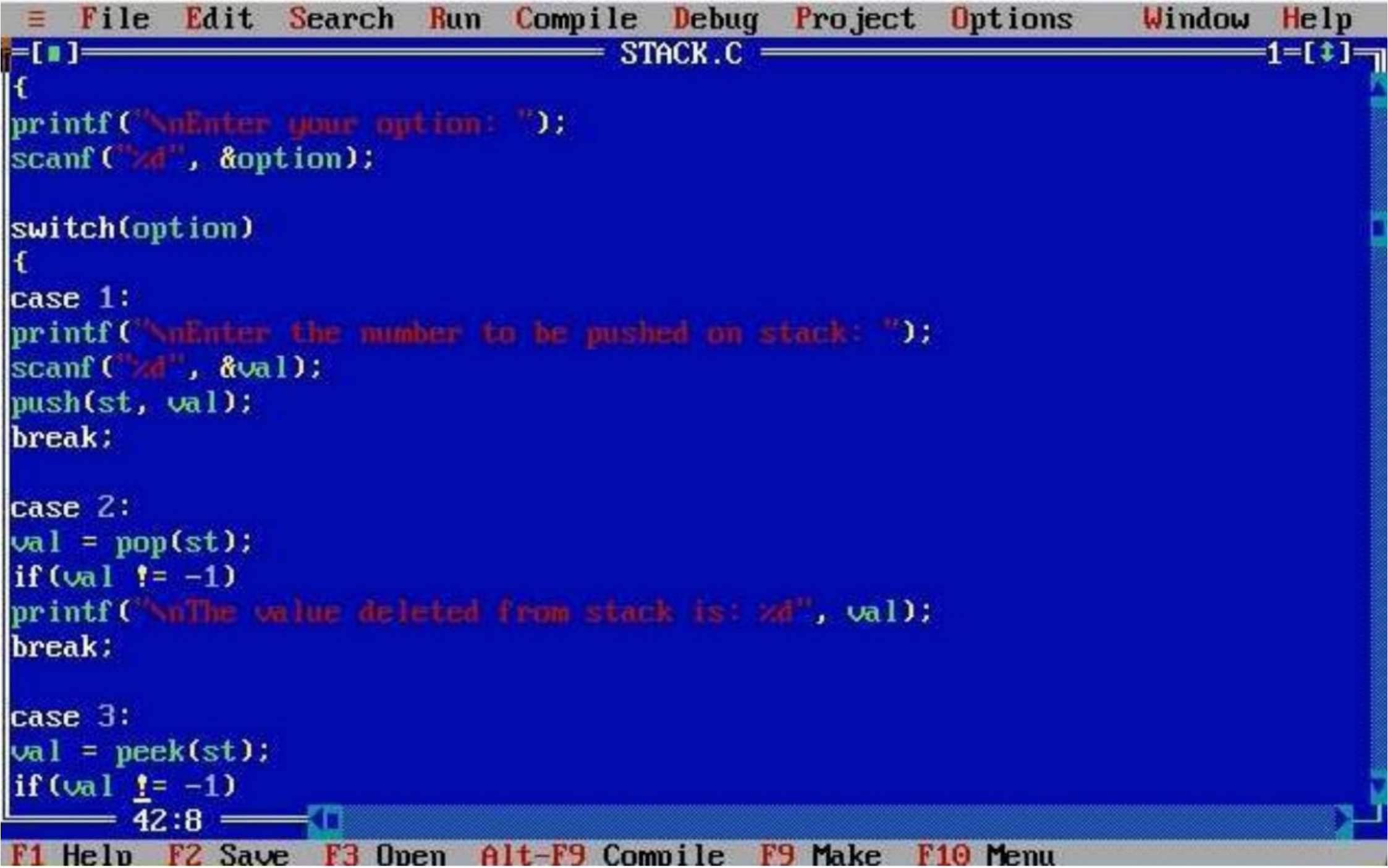

STACK.C
#include <stdio.h>
#include <conio.h>
#include <stdlib.h>
#define MAX 3
int st[MAX], top = -1;
void push(int st[], int val);
int pop(int st[]);
int peek(int st[]);
void display(int st[]);
void main()
{
int val, option;
printf("\n****** MAIN MENU ******");
printf("\n1. PUSH");
printf("\n2. POP");
printf("\n3. PEEK");
printf("\n4. DISPLAY");
printf("\n5. EXIT");
do
1:1
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window Help
STACK.C
{
printf("\nEnter your option: ");
scanf("%d", &option);
switch(option)
{
case 1:
printf("\nEnter the number to be pushed on stack: ");
scanf("%d", &val);
push(st, val);
break;
case 2:
val = pop(st);
if(val != -1)
printf("\nThe value deleted from stack is: %d", val);
break;
case 3:
val = peek(st);
if(val != -1)
42:8
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

```
≡ File  Edit  Search  Run  Compile  Debug  Project  Options    Window  Help
[■]                          STACK.C                                  1=[↕]
printf("\nThe value stored at top of stack is: %d", val);
break;

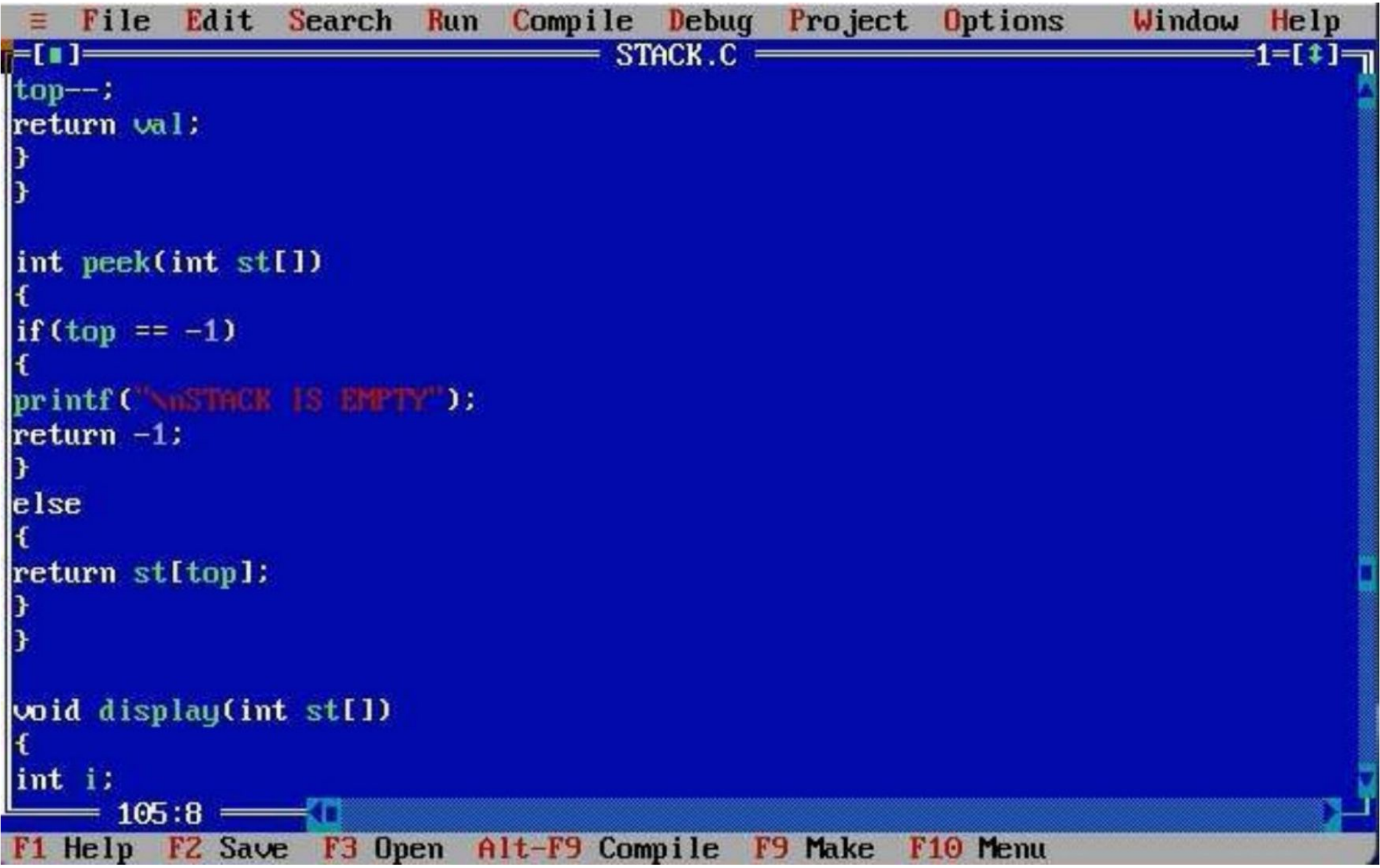
case 4:
display(st);
break;

case 5:
printf("\nExiting...");
break;

default:
printf("\nInvalid Choice");
}
getch();
} while(option != 5);
}

void push(int st[], int val)
{
if(top == MAX - 1)
        63:8
F1 Help  F2 Save  F3 Open  Alt-F9 Compile  F9 Make  F10 Menu
```


File Edit Search Run Compile Debug Project Options Window Help
STACK.C
{
printf("\nSTACK OVERFLOW");
}
else
{
top++;
st[top] = val;
}
}
int pop(int st[])
{
int val;
if(top == -1)
{
printf("\nSTACK UNDERFLOW");
return -1;
}
else
{
val = st[top];
84:8
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window Help
STACK.C
top--;
return val;
}
}
int peek(int st[])
{
if(top == -1)
{
printf("\nSTACK IS EMPTY");
return -1;
}
else
{
return st[top];
}
}
void display(int st[])
{
int i;
105:8
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu


File Edit Search Run Compile Debug Project Options Window Help
STACK.C
{
return st[top];
}
}
void display(int st[])
{
int i;
if(top == -1)
{
printf("\nSTACK IS EMPTY");
}
else
{
printf("\nStack elements are:\n");
for(i = top; i >= 0; i--)
{
printf("%d\n", st[i]);
}
}
}
118:8
F1 Help F2 Save F3 Open Alt-F9 Compile F9 Make F10 Menu

```
C:\TURBOC3\BIN>TC

****** MAIN MENU ******
1. PUSH
2. POP
3. PEEK
4. DISPLAY
5. EXIT
Enter your option: 1

Enter the number to be pushed on stack: 10

Enter your option: 1

Enter the number to be pushed on stack: 20

Enter your option: 4

Stack elements are:
20
10
```